UNIVERSITY OF HELSINKI

NETWORK ANALYSIS

HOMEWORK

EXERCISE 2

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PROBLEM 1

Visualization Fig.1 shows the visualization of graph *barn-swallow*.

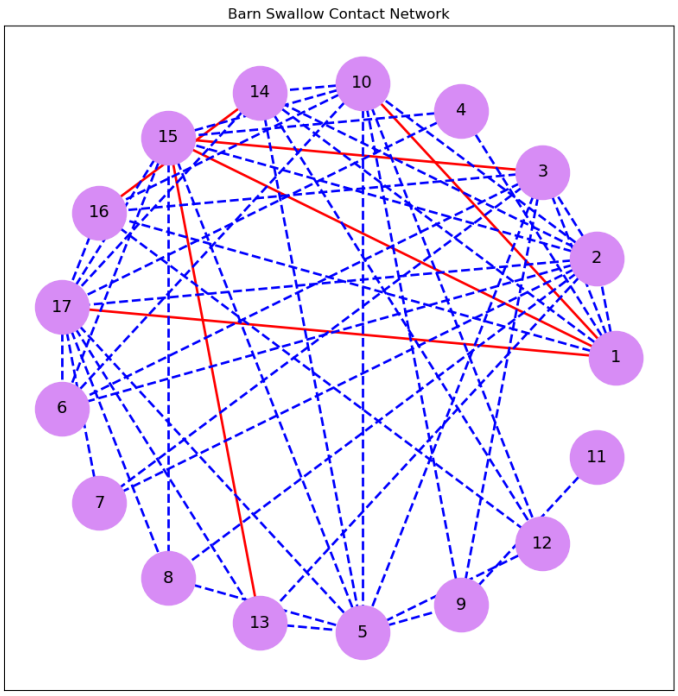

Figure 1: *barn-swallow* graph visualization.

STCP Violation STCP satisfying nodes: $[2, 3, 4, 10, 14, 16, 17, 6, 7, 8, 13, 5, 9, 12, 11]$.
For this *barn-swallow* network, I see that STCP satisfying nodes have at most **1** strong ties only, which we don't have to check whether there is a tie between neighbors of each satisfying node.

Problem 2

Balance in Signed Fig.2 shows why and where the criteria is not satisfied and Fig.3 shows 2 disjoint sets of nodes.

```
[ 14:24:25 ] → python hw2_p2.py
There are 7 supernodes and all are valid.
2024-01-30 14:24:27.717 | INFO     | __main__:bfs_check:268 - Found Odd Length Cycle at: ('s4', 's3')
Odd cycle found in reduced graph
Simple Graph is Unbalanced
```

Figure 2: The result as running the script shows why and where the criteria is not satisfied.

```
There are 6 supernodes and all are valid.
Bipartite mapping of modified simple graph nodes
Group X :  [1, 2, 3, 4, 5, 11, 15]
Group Y:  [8, 6, 7, 9, 10, 12, 13, 14]
```

Figure 3: The result as running the script shows 2 disjoint set of nodes..

Asymmetric Edges Fig.5 shows the number of asymmetric node pairs the result of checking balance as running with the implemented algorithm. Fig.4 shows the histogram of the absolute differences.

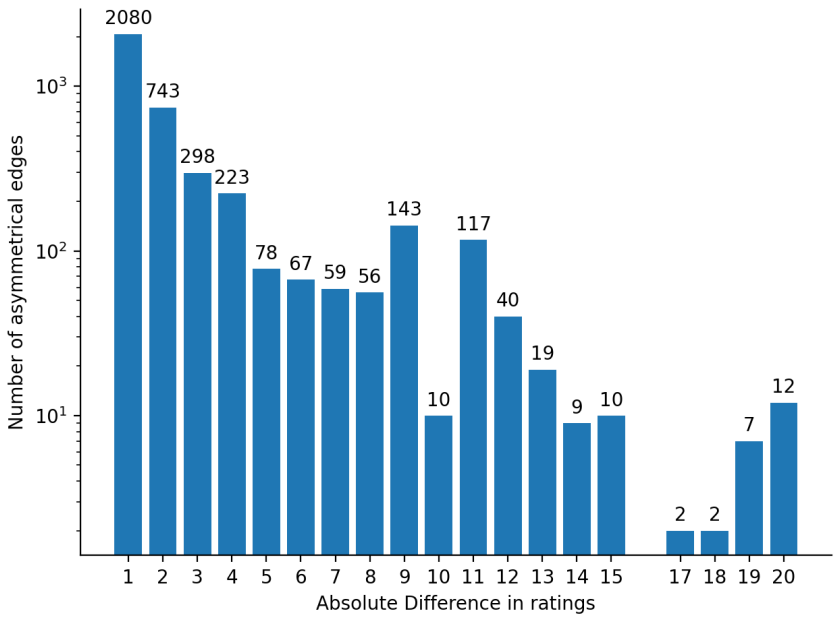

Figure 4: Histogram of rating's absolute difference.

```
There are 3975 pairs of nodes that have asymmetric edges
There are 11 supernodes and all are valid.
Bitcoin Network is balanced
```

Figure 5: The result as running the script shows 2 disjoint set of nodes..